

PWU-AB-1 / PWU-AB-3 — A/B video evidence recordings

Revision: 1 **Last modified:** 2026-06-19T01:30:00Z **Status:** active **Type:** Task — evidence collection **Scope:** Recorded asciinema casts of the real U-Boot 2024.01 + QEMU virt + TCG console output, converted to MP4 + GIF, for the A/B slot-switch and auto-rollback demos.

Recording files

File	Size	Format	Content
helix_ota-ab-slot-switch.mp4	1.3 MB	MP4 (h.264)	Two sequential boots: slot A then slot B
helix_ota-ab-slot-switch.cast	55 KB	asciinema v3	Raw cast of slot switch demo
helix_ota-ab-rollback.mp4	1.4 MB	MP4 (h.264)	Two boots: ROLLBACK then CONTROL
helix_ota-ab-rollback.cast	60 KB	asciinema v3	Raw cast of rollback demo

Recording 1: A/B Slot Switch (PWU-AB-1)

Console key frames

Boot A: BOOT_ORDER="A B"

```
=> setenv BOOT_ORDER A B
=> printenv BOOT_ORDER -> BOOT_ORDER=A B
=> source boot.scr
A/B: active_slot=A root=/dev/vda2
# echo HELIX_SLOTID=$(cat /etc/slot_id) -> HELIX_SLOTID=A
# echo HELIX_ROOTDEV=$(findmnt -no SOURCE /) -> HELIX_ROOTDEV=/dev/vda2
```

Boot B: BOOT_ORDER="B A"

```
=> setenv BOOT_ORDER B A
=> printenv BOOT_ORDER -> BOOT_ORDER=B A
=> source boot.scr
A/B: active_slot=B root=/dev/vda3
# echo HELIX_SLOTID=$(cat /etc/slot_id) -> HELIX_SLOTID=B
# echo HELIX_ROOTDEV=$(findmnt -no SOURCE /) -> HELIX_ROOTDEV=/
dev/vda3
```

Recording 2: Auto-Rollback (PWU-AB-3)

Console key frames

ROLLBACK: BOOT_ORDER="B A", bootcount=2 > bootlimit=1

```
=> setenv bootcount 2, bootlimit 1, upgrade_available 1
=> source boot.scr
A/B: bootcount=2 > bootlimit=1 -> rolling back (altbootcmd swap)
Saving Environment to Flash...
A/B: BOOT_ORDER=A B active_slot=A root=/dev/vda2
# echo HELIX_SLOTID=A
```

CONTROL: BOOT_ORDER="B A", bootcount=1 (not exhausted)

```
=> setenv bootcount 1, bootlimit 1, upgrade_available 1
=> source boot.scr
A/B: BOOT_ORDER=B A active_slot=B root=/dev/vda3
# echo HELIX_SLOTID=B
```

Summary

Feature	Verdict	Key evidence
Slot A boot	PASS	BOOT_ORDER="A B" -> root=/dev/vda2 -> slot_id=A
Slot B boot	PASS	BOOT_ORDER="B A" -> root=/dev/vda3 -> slot_id=B
Auto-rollback	PASS	bc=2 > bl=1 -> guard fires -> swaps to slot A
Control (no rollback)	PASS	bc=1 = bl=1 -> boots head B directly